

Welcome to SYST38767

Objectives

1. Course objectives and overview
2. Café business case introduction
3. Roles in cloud computing

Activities

AWS Documentation Scavenger Hunt

Course objectives

- Recall cloud computing services and models
- Describe developing on AWS
- Write code that interacts with Amazon Simple Storage Service (Amazon S3) by using AWS SDKs
- Explain the role of AWS Identity and Access Management (IAM)
- Write code that interacts with Amazon DynamoDB by using AWS SDKs
- Create a REST API by using Amazon API Gateway
- Create functions with AWS Lambda by using AWS SDKs
- Configure containers
- Explain caching with Amazon CloudFront and Amazon ElastiCache
- Develop solutions with Amazon Simple Queue Service (Amazon SQS) and Amazon Simple Notification Service (Amazon SNS)
- Terraform

Course Content – 7A

- Week 1: Review TELE20483
- Week 2: Exploring Cloudshell
- Week 3: Developing Storage Solutions - S3
- Week 4: Developing Flexible NoSQL Solutions - DynamoDB
- Week 5: Developing REST APIs
- Week 6: Review
- Week 7 : Midterm

Week 2: Introduction to Developing on AWS

Module sections

1. Introduction
2. Systems development lifecycle
3. Steps to get started developing on AWS
4. Fundamentals of working with the AWS SDKs

Module lab

Exploring AWS CloudShell and an IDE



Week 3: Developing Storage Solutions

Module sections

1. Introduction
2. Introducing Amazon S3
3. Creating S3 buckets
4. Working with S3 objects
5. Protecting data and managing access to Amazon S3 resources

Module lab

Working with Amazon S3



Week 4: Developing Flexible NoSQL Solutions

Module sections

1. Introduction
2. Introducing AWS database options
3. Key concepts for DynamoDB
4. Partitions and data distribution
5. Secondary indexes
6. Read/write throughput
7. Streams and global tables
8. Backup and restore
9. Basic operations for DynamoDB tables



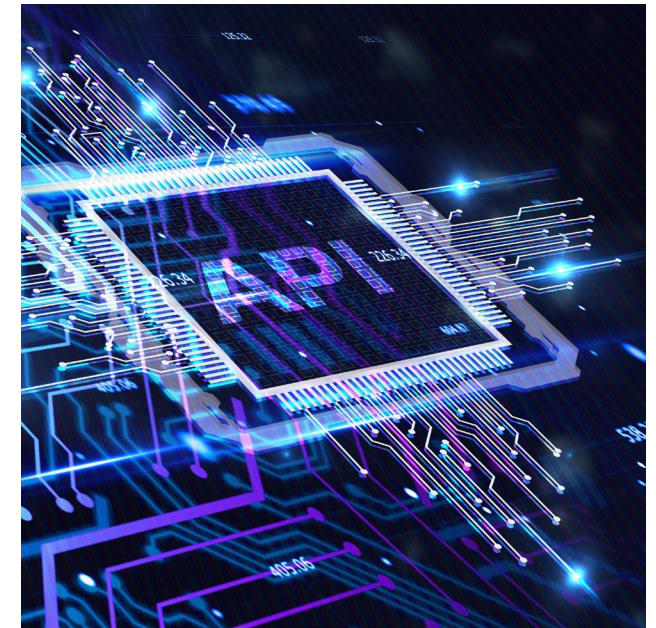
Module lab

Working with DynamoDB

Week 5: Developing REST APIs

Module sections

1. Introduction
2. Introducing APIs
3. Introducing Amazon API Gateway
4. Creating a REST API
5. Integrating with API Gateway
6. Deploying an API
7. Controlling access to a REST API
8. Monitoring a REST API
9. Optimizing API Gateway



Module lab

Developing REST APIs with API Gateway



Café business Introduction

Café and bakery



Café owners and staff

Frank

- Co-owner of café
- Retired from the Navy
- Likes to bake
- Not technical



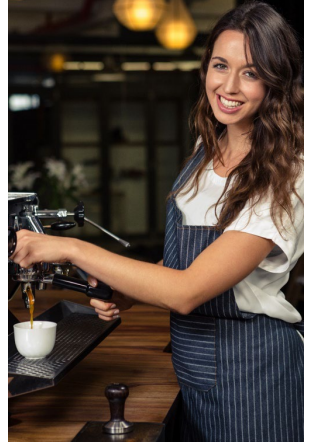
Martha

- Co-owner of café
- Retired accountant
- Knows how to use spreadsheets, otherwise not technical



Sofía

- Daughter of Frank and Martha
- Manages the café's supply chain
- Technical skills, including programming, future business administration student
- Started to use AWS



Nikhil

- Café employee, visual design skills
- Interested in learning cloud computing
- Might take on more responsibilities at the café when Sofía starts her studies at the university



AWS consultants, café visitors

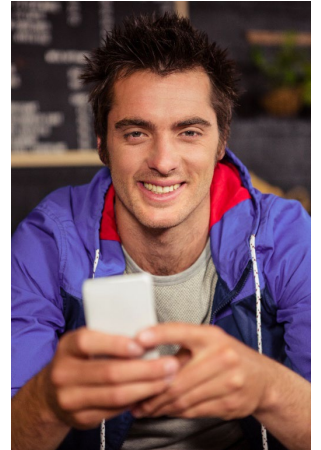
Olivia

- An AWS solutions architect
- Technical, with specialties in databases and network technologies



Mateo

- Systems administrator and engineer
- Likes to find ways to automate and create repeatable solutions
- Knows the importance of backups and disaster recovery in solution design



Faythe

- Developer, experienced with AWS programming interfaces
- Knowledgeable about cloud security



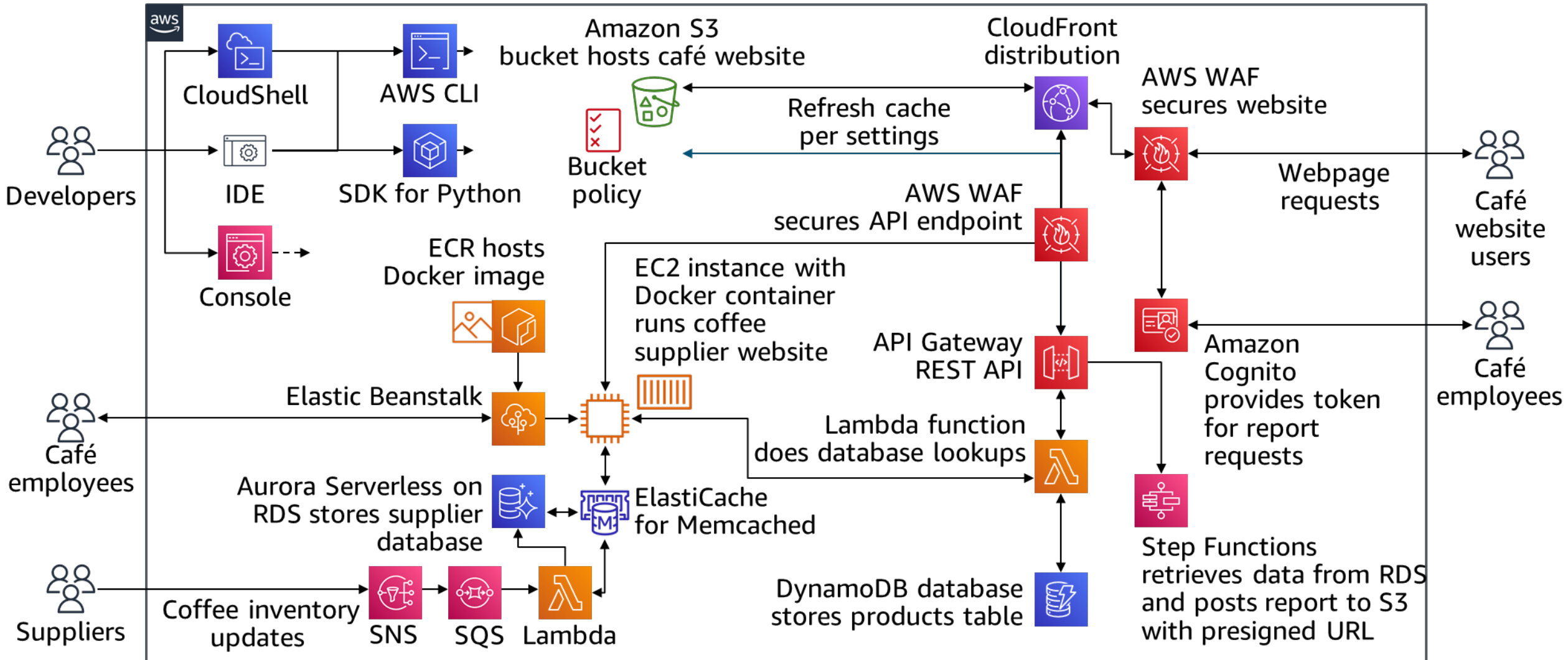
You will build solutions for the café in the labs in this course

The café has business needs that can be solved with cloud development.



The café employees and the consultants often socialize and share cloud architecture ideas.

The café application that you will build





Roles in cloud computing

Roles in cloud computing: IT professional



Characteristics

- Generalist, might manage an application
- Often manages a production environment
- Highly technical
- Might have significant or limited experience in cloud technologies
- Might specialize in one area (such as security or storage)

Job titles: IT administrator, systems administrator, network administrator

Roles in cloud computing: IT leader



Characteristics

- Leads a team of IT professionals
- Responsible for day-to-day operations
- Manages a budget, and stays informed about and chooses new technologies
- Hands on during early stages of a project, then delegates the team to take over

Job titles: IT manager, IT director, IT supervisor

Roles in cloud computing: DevOps engineer



Characteristics

- Builds out the infrastructure that applications run on, often in the cloud
- Follows the guidelines of the cloud architect
- Prefers experimenting and trying things out rather than a lot of reading

Job titles: DevOps engineer, reliability engineer, build engineer

Roles in cloud computing: Cloud architect



Characteristics

- Stays up to date with new technologies and helps decide which to use
- Provides documentation, processes, and tooling to developers
- Gives developers freedom to innovate
- Common challenges include:
 - Resource management
 - Cost optimization
 - Defining best practices for performance, reliability, and security

Job titles: cloud architect, systems engineer, systems analyst

Roles in cloud computing: Developer



Characteristics

- Writes, tests, and fixes code
- Thinks about projects at the application level
- Likes sample code
- Works with APIs and SDKs

Job titles: software developer, systems architect, software development manager

Activity: AWS Documentation Scavenger Hunt



AWS Documentation Scavenger Hunt

1. Which section of the website contains links to the supported AWS SDKs and toolkits?
2. How would you navigate to the method for creating an S3 bucket in the AWS SDK for Python (Boto3)?
3. How would you navigate to the Getting Started topic in the AWS Documentation for IAM?
4. Which section of the website has a tutorial on how to build a serverless application?
5. Which section of the website provides a link to AWS technical whitepapers?

<https://docs.aws.amazon.com>

Module summary

In summary, in this module, you learned to:

- Identify course prerequisites and objectives
- Recognize the café business case
- Identify roles in cloud computing
- Familiarize yourself with AWS documentation



Thank you

Corrections, feedback, or other questions?

Contact us at <https://support.aws.amazon.com/#/contacts/aws-academy>.